

XRPL Validator — Daily Engineering Report • 2026-06-14

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- 7 commits (xrpl-validator 7)

Engineering changes

- **fix validator: remove dead full-DB sweep footgun (reaudit F2)** (a99dbb1 , xrpl-validator) start_sweep + fetch_batch re-fetched every state.rocks key at one pinned ledger and wrote each with an unconditional db.put NOT paired with a hasher update. Run alongside live sync it regressed freshly-synced keys to stale pinned-seq values while the in-memory hasher kept the live value — silent rocks rot that detonates on the next restart (hasher rebuilt from rocks → wrong account_hash → 3 strikes → halt). Zero ...
- **fix validator: refuse signing startup with untrusted RPC endpoints (reaudit F7)** (7501eab , xrpl-validator) A signing validator on a host with no local rippled (the validator host) silently fails over from localhost to the public clusters (xrplcluster.com / s1.ripple.com) when XRPL_RPC_URL/XRPL_WS_URL are unset — a consensus-divergence source and a violation of the never-hammer-public-RPCs rule. The binary never enforced the operational 'MUST set both env vars' rule.
- **feat validator: encode crash-consistency contract via clean-shutdown marker (reaudit F5)** (28f6658 , xrpl-validator) The 'wipe state.rocks + bulk-resync on unclean restart' contract lived only in operator discipline. A clean shutdown now writes clean_shutdown.marker (SIGTERM handler, after the leaf-cache save); boot consumes it. Marker present => prior shutdown clean, state consistent, resume. Marker absent while state.rocks exists => unclean (OOM/SIGKILL/power) => state may be torn (hasher rebuilds from inconsistent rocks → ...
- **feat validator: refresh active amendments on flag ledgers (reaudit F6/va-07)** (ecf3ebc , xrpl-validator) The FFI verifier fetched the active mainnet amendment set once at startup and froze it for the process lifetime. An amendment that reaches 2-week majority and activates while the validator runs would be absent → every tx exercising it diverges → 3 strikes → halt; this fires more often in the post-3.2.0 wave.
- **fix ledger: preserve leaf data on SHAMap delete collapse (reaudit F4 / val-110)** (826406e , xrpl-validator) delete_from_mut's structural-collapse path promoted the surviving leaf via find_sole_leaf -> LeafNode::new_hash_only(key, hash), which zeroes data (data: Vec::new()). In a State tree the surviving sibling kept its hash but lost its bytes — invisible to the hash-based signing path, but lookup() returned Some([]). This blocked lockstep M1 and any path trusting SHAMap::delete.

Docs and operations

- **chore validator: commit ffi_test sidecar arg drift to extended apply_ledger_in_order signature (reaudit F8a)** (de6ea4b , xrpl-validator)

- `style validator`: use `u32::is_multiple_of` in `is_flag_ledger` (clippy) (2279891 , `xrpl-validator`)

Appendix — commits

Time	Repo	Commit	Subject
09:10	xrpl-validator	de6ea4b	chore validator: commit ffi_test sidecar arg drift to extended apply_ledger_in_order signature (reaudit F8a)
09:21	xrpl-validator	a99dbb1	fix validator: remove dead full-DB sweep footgun (reaudit F2)
09:28	xrpl-validator	7501eab	fix validator: refuse signing startup with untrusted RPC endpoints (reaudit F7)
09:34	xrpl-validator	28f6658	feat validator: encode crash-consistency contract via clean-shutdown marker (reaudit F5)
09:47	xrpl-validator	ecf3ebc	feat validator: refresh active amendments on flag ledgers (reaudit F6/va-07)
09:53	xrpl-validator	826406e	fix ledger: preserve leaf data on SHAMap delete collapse (reaudit F4 / val-110)
10:10	xrpl-validator	2279891	style validator: use <code>u32::is_multiple_of</code> in <code>is_flag_ledger</code> (clippy)